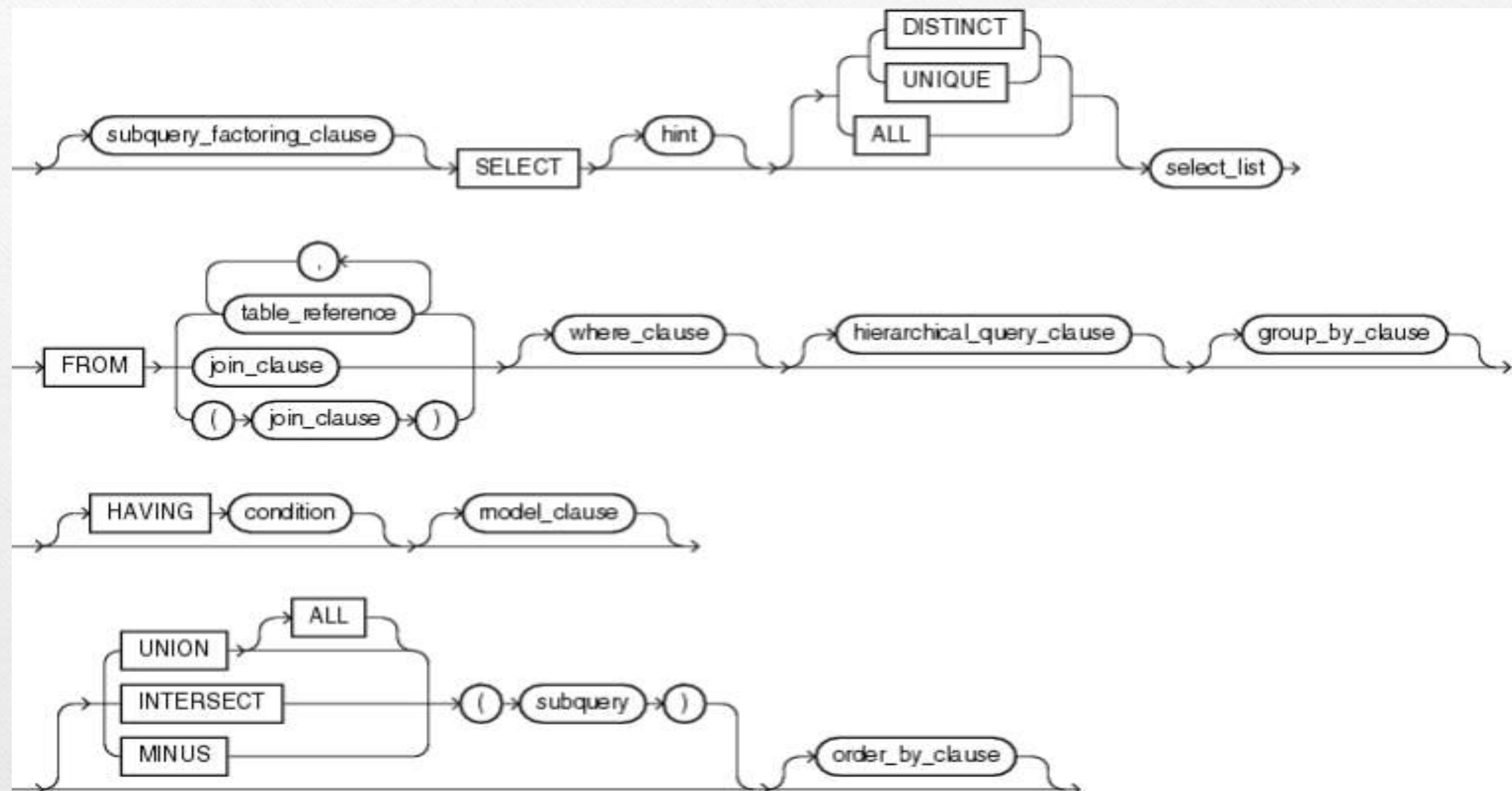


СУБД

Лекция 5

Выборка данных



Выборка данных из одной таблицы

- SELECT имена_полей
 - FROM название_таблицы;
-
- SELECT *
 - FROM название_таблицы;

ROWID

- Select rowid, column1
- From tab1

Арифметические действия над датами

```
SELECT TO_DATE('31.07.2015')+2
```

```
FROM dual;
```

02.08.2015

```
SELECT TO_DATE(02.08.2015)-TO_DATE  
( '31.07.2015')
```

```
FROM dual
```

2

Конкатенация строк

```
SELECT CONCAT('Hello ', 'world')  
FROM DUAL;
```

```
SELECT 'Hello' || ' || 'world'  
FROM dual
```

NVL

- SELECT first_name, last_name,
nvl(middle_name,'')
- FROM persons

Case

- Case
 - входное значение
- WHEN условие THEN результат
- ...
- END

Символьные функции

Функция	Описание
Lower	Приведение к нижнему регистру
Upper	Приведение к upper регистру
Initcap	Первая буква заглавная
Concat	конкатенация
Substr	Выделение подстроки
Length	Длина строки
Instr	Поиск подстроки в строке
Lpad rpad	Добавление символов слева справа
Trim	усечение
replace	Замена подстроки

DISTINCT

`select manager_id from employees;`

AZ	MANAGER_ID
	124
	124
	101
	100
	201
	101
	101
	101
	205

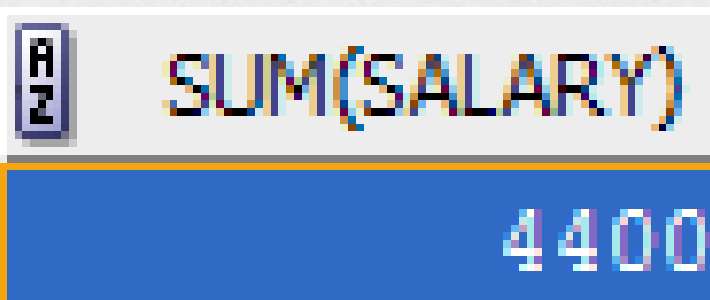
`select distinct manager_id from employees;`

AZ	MANAGER_ID
	100
	(null)
	123
	120
	121
	147
	205
	108
	148

Агрегатные функции

Функция	Описание
AVG	Среднее значение столбца
Count	Количество строк
Min	Минимальное значение
Max	Максимальное значение
Sum	Сумма

select sum (salary) from employees where department_id=10;



A screenshot of a SQL query result. The query is `SUM(SALARY)`. The result is displayed in a blue box with the value `4400`.

Группировки

- SELECT count(employee_id), department_id
- FROM departments
- GROUP BY department_id

Объединение таблиц

- Горизонтальное объединение
- Вертикальное объединение

Natural join

- Объединение по одинаковым названиям.

Column Name	Data Type
DEPARTMENT_ID	NUMBER (4)
DEPARTMENT_NAME	VARCHAR2 (30 Byte)
MANAGER_ID	NUMBER (6)
LOCATION_ID	NUMBER (4)

Column Name	Data Type
LOCATION_ID	NUMBER (4)
STREET_ADDRESS	VARCHAR2 (40 Byte)
POSTAL_CODE	VARCHAR2 (12 Byte)
CITY	VARCHAR2 (30 Byte)
STATE_PROVINCE	VARCHAR2 (25 Byte)
COUNTRY_ID	CHAR (2 Byte)

SELECT * FROM departments NATURAL JOIN locations;

LOCATION_ID	DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	STREET_ADDRESS	POSTAL_CODE	CITY	STATE_PROVINCE	COUNTRY_ID
1400	60	IT	103	2014 Jabberwocky Rd	26192	Southlake	Texas	US
1500	50	Shipping	121	2011 Interiors Blvd	99236	South San Francisco	California	US
1700	10	Administration	200	2004 Charade Rd	98199	Seattle	Washington	US
1700	30	Purchasing	114	2004 Charade Rd	98199	Seattle	Washington	US
1700	90	Executive	100	2004 Charade Rd	98199	Seattle	Washington	US
1700	100	Finance	108	2004 Charade Rd	98199	Seattle	Washington	US
1700	110	Accounting	205	2004 Charade Rd	98199	Seattle	Washington	US
1700	120	Treasury		2004 Charade Rd	98199	Seattle	Washington	US
1700	130	Corporate Tax		2004 Charade Rd	98199	Seattle	Washington	US
1700	140	Control And Credit		2004 Charade Rd	98199	Seattle	Washington	US

Объединение с использованием USING

- SELECT *
- FROM departments JOIN locations USING
(location_id)

Объединение с использованием on

- SELECT *
- FROM departments d JOIN locations l ON (d.location_id > l.location_id)

DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID	LOCATION_ID_1	STREET_ADDRESS	POSTAL_CODE	CITY	STATE_PROVINCE	COUNTRY_ID
60	IT	103	1400	1000	1297 Via Cola di Rie	00989	Roma		IT
50	Shipping	121	1500	1000	1297 Via Cola di Rie	00989	Roma		IT
10	Administration	200	1700	1000	1297 Via Cola di Rie	00989	Roma		IT
30	Purchasing	114	1700	1000	1297 Via Cola di Rie	00989	Roma		IT
90	Executive	100	1700	1000	1297 Via Cola di Rie	00989	Roma		IT
100	Finance	108	1700	1000	1297 Via Cola di Rie	00989	Roma		IT
110	Accounting	205	1700	1000	1297 Via Cola di Rie	00989	Roma		IT
120	Treasury		1700	1000	1297 Via Cola di Rie	00989	Roma		IT
130	Corporate Tax		1700	1000	1297 Via Cola di Rie	00989	Roma		IT
140	Control And Credit		1700	1000	1297 Via Cola di Rie	00989	Roma		IT
150	Shareholder Services		1700	1000	1297 Via Cola di Rie	00989	Roma		IT
160	Benefits		1700	1000	1297 Via Cola di Rie	00989	Roma		IT
170	Manufacturing		1700	1000	1297 Via Cola di Rie	00989	Roma		IT
180	Construction		1700	1000	1297 Via Cola di Rie	00989	Roma		IT

-
- SELECT *
 - FROM departments d, locations l
 - WHERE d.location_id > l.location_id

Self join

```
SELECT
    E1.LAST_NAME,
    E2.LAST_NAME
FROM employees e1
    JOIN employees e2
    ON
    (E1.MANAGER_ID =
    E2.EMPLOYEE_ID)
```

LAST_NAME	LAST_NAME_1
OConnell	Mourgos
Grant	Mourgos
Whalen	Kochhar
Hartstein	King
Fay	Hartstein
Mavris	Kochhar
Baer	Kochhar
Higgins	Kochhar
Gietz	Higgins
Kochhar	King
De Haan	King
Hunold	De Haan
Ernst	Hunold
Austin	Hunold
Pataballa	Hunold
Lorentz	Hunold

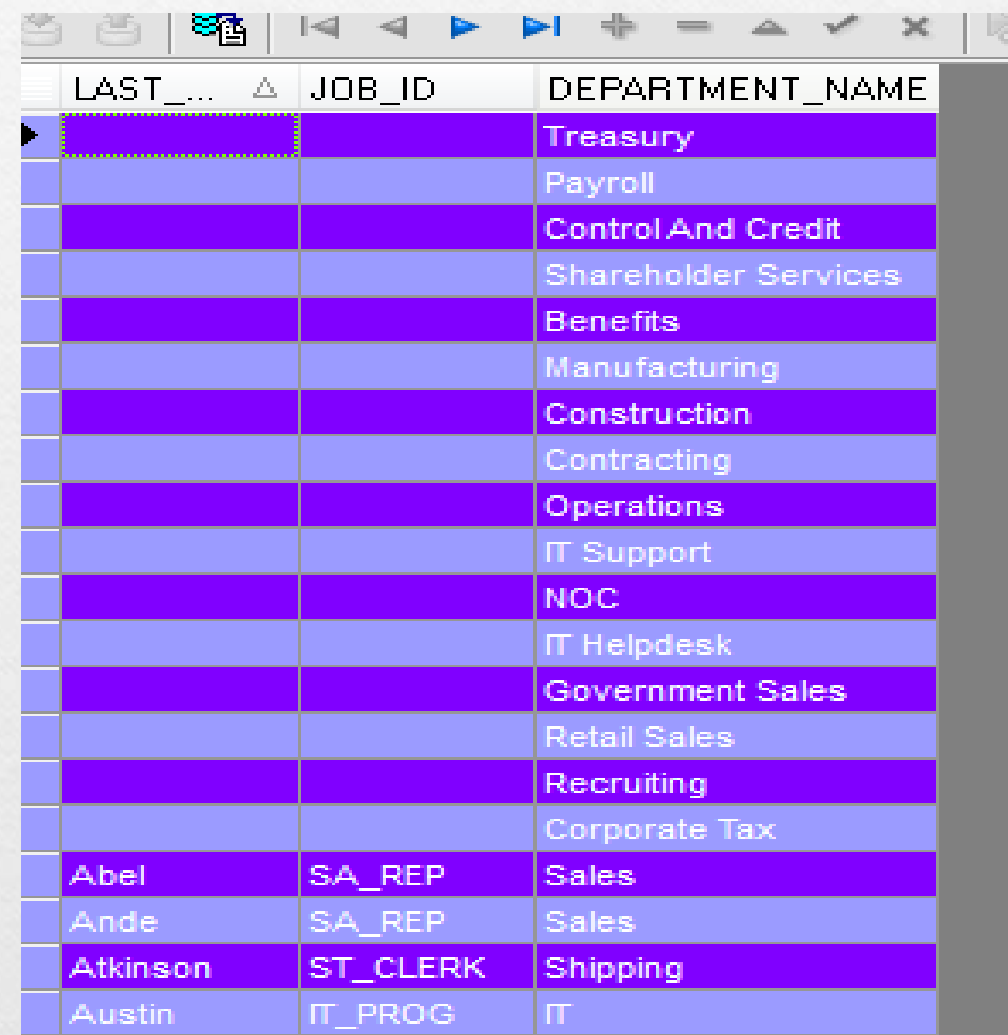
Outer join

```
SELECT
    e.Last_name,
    e.job_id,
    d.department_name
FROM employees e
    LEFT OUTER JOIN
departments d ON
(E.DEPARTMENT_ID =
D.DEPARTMENT_ID)
```

LAST_NAME	JOB_ID	DEPARTMENT_...
Grant	SA_REP	
Gietz	AC_ACCOUNT	Accounting
Higgins	AC_MGR	Accounting
Whalen	AD_ASST	Administration
De Haan	AD_VP	Executive
Kochhar	AD_VP	Executive
King	AD PRES	Executive
Urman	FI_ACCOUNT	Finance
Greenberg	FI_MGR	Finance
Faviet	FI_ACCOUNT	Finance
Chen	FI_ACCOUNT	Finance
Popp	FI_ACCOUNT	Finance
Sciarra	FI_ACCOUNT	Finance
Mavris	HR_REP	Human Resources
Lorentz	IT_PROG	IT

Outer join

- SELECT e.Last_name,
e.job_id,
d.department_name
- FROM employees e
- **RIGHT** OUTER
JOIN departments d ON
(E.DEPARTMENT_ID =
D.DEPARTMENT_ID)



The screenshot shows a database query result window with a toolbar at the top. The table has three columns: LAST_NAME, JOB_ID, and DEPARTMENT_NAME. The data is as follows:

LAST_NAME	JOB_ID	DEPARTMENT_NAME
		Treasury
		Payroll
		Control And Credit
		Shareholder Services
		Benefits
		Manufacturing
		Construction
		Contracting
		Operations
		IT Support
		NOC
		IT Helpdesk
		Government Sales
		Retail Sales
		Recruiting
		Corporate Tax
Abel	SA_REP	Sales
Ande	SA_REP	Sales
Atkinson	ST_CLERK	Shipping
Austin	IT_PROG	IT

Outer join

```
SELECT e.Last_name,  
e.job_id,  
d.department_name
```

```
FROM employees e
```

```
  FULL OUTER JOIN  
departments d ON  
(E.DEPARTMENT_ID =  
D.DEPARTMENT_ID)
```

LAST_NAME	JOB_ID	DEPARTMENT_...
Grant	SA_REP	
Gietz	AC_ACCOUNT	Accounting
Higgins	AC_MGR	Accounting
Whalen	AD_ASST	Administration
		Benefits
		Construction
		Contracting
		Control And Credit
		Corporate Tax
De Haan	AD_VP	Executive
King	AD PRES	Executive
Kochhar	AD_VP	Executive
Popp	FI_ACCOUNT	Finance
Urman	FI_ACCOUNT	Finance
Sciarra	FI_ACCOUNT	Finance
Greenberg	FI_MGR	Finance
Faviet	FI_ACCOUNT	Finance

Cross join

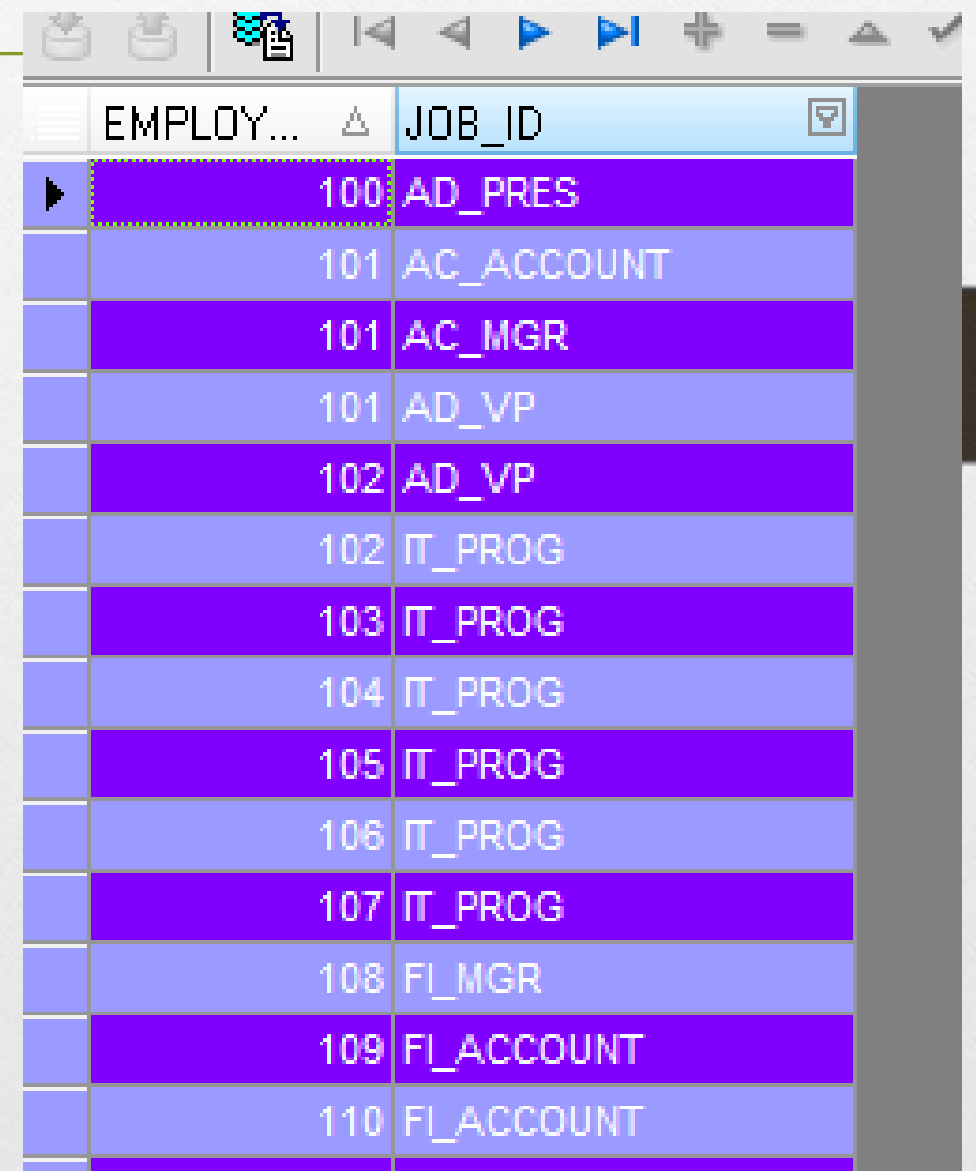
- SELECT e.Last_name, e.job_id,
d.department_name
- FROM employees e
- CROSS JOIN departments d

Cross join

```
SELECT e.Last_name, e.job_id, d.department_name  
FROM employees e, departments d
```

Union

- select employee_id, job_id
- from employees
- union
- select employee_id, job_id
- from job_history



The screenshot shows a database query result window with a toolbar at the top. The table has two columns: 'EMPLOY...' and 'JOB_ID'. The data is a union of two queries, resulting in 11 rows. The first row is highlighted with a mouse cursor.

EMPLOY...	JOB_ID
100	AD_PRES
101	AC_ACCOUNT
101	AC_MGR
101	AD_VP
102	AD_VP
102	IT_PROG
103	IT_PROG
104	IT_PROG
105	IT_PROG
106	IT_PROG
107	IT_PROG
108	FL_MGR
109	FL_ACCOUNT
110	FL_ACCOUNT

Union all

- select employee_id, job_id
- from employees
- Union All
- select employee_id, job_id
- from job_history

intersect

- select employee_id, job_id
- from employees
- intersect
- select employee_id, job_id
- from job_history

minus

- select employee_id, job_id
- from employees
- minus
- select employee_id, job_id
- from job_history

Connect by

- **SELECT LEVEL NAME, ID, PARENT_ID**
 - **FROM EMPL**
 - **CONNECT BY PRIOR ID = PARENT_ID;**
- **SELECT NAME, ID, PARENT_ID**
 - **FROM EMPL**
 - **CONNECT BY PRIOR ID = PARENT_ID**
 - **START WITH ID IN (SELECT ID FROM EMPL WHERE PARENT_ID IS NULL);**

-
- **SELECT NAME, ID, PARENT_ID, LEVEL**
 - **FROM EMPL**
 - **CONNECT BY PRIOR ID = PARENT_ID**
 - **START WITH ID = 1;**

